



Asymmetric cost pass-through? Empirical evidence on the role of market power, search and menu costs



Jens-Peter Loy^{a,*}, Christoph R. Weiss^b, Thomas Glauben^c

^a Department of Agricultural Economics, Christian-Albrechts-University, Kiel, Germany

^b Vienna University of Economics and Business (WU), Austria

^c Leibniz Institute of Agricultural Development in Central and Eastern Europe, Halle an der Saale, Germany

ARTICLE INFO

Article history:

Received 20 October 2014

Received in revised form 18 January 2016

Accepted 24 January 2016

Available online 3 February 2016

JEL classification:

C32

D21

L11

L81

Keywords:

Asymmetric cost pass-through

Market power

Menu costs

Search costs

Milk

Food retailing

ABSTRACT

We estimate cost pass-through rates based on data for store-level retail prices and whole-sale costs for up to 90 different brands of milk sold in 327 retail stores across Germany. The abundance of the data allows us to identify heterogeneity in pass-through across retailers and relate it to underlying explanatory factors such as retailer market power, measures of consumer search and menu costs. Results from a threshold-error-correction-process clearly provide empirical support for the 'rockets and feathers' phenomenon: prices increase more quickly than they decrease. In contrast to much of the existing literature, which explains the 'rockets and feathers' phenomenon as a result of retailers exercising market power, we find the opposite result: the degree of asymmetry in prices is negatively related to market power (decreases with the Lerner-Index). Our measure of (cross-product) search costs does not account for observable differences in asymmetric cost pass-through rates between products. The 'rockets and feathers' observation is better explained by product-specific differences in menu costs ('mistake costs') and costs associated with stock-outs which can lead to a loss of consumer goodwill.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

An asymmetric adjustment of prices to cost increases and decreases (asymmetric cost pass-through, 'aCPT')¹ has been observed for many agricultural and energy markets (see Frey and Manera (2007), Eckert (2013) or Perdiguer-Garcia (2013) for recent surveys of the field). Different explanations for an aCPT have been proposed in the existing literature: non-competitive markets (market power), costs of price adjustment (menu costs and 'mistake costs'), product-specific differences in costs of stock-outs, and search costs.

While there is ample empirical evidence on incomplete and asymmetric price transmission, the existing studies do not provide reliable support to interpret this as evidence in favor of one of the different explanations offered in the literature. A large share of this literature focuses on one product market using only time series data and implicitly assumes competition

* Corresponding author. Tel.: +49 1704520158.

E-mail address: jploy@ae.uni-kiel.de (J.-P. Loy).

¹ The existing literature uses the terms 'vertical price transmission' and 'cost-pass-through' (CPT) interchangeably to characterize the impact of factor price (cost) changes on downstream product prices (Misra and Khan, 2010). We will use the term 'cost-pass-through' (CPT) or 'asymmetric cost-pass-through' (aCPT) in the present analysis.

within this market to be global (some exceptions are mentioned in the next section). ‘Unless important changes in market power are known to have occurred within the study period, this sort of analysis provides no basis for comparing price transmission under conditions of more and less market power because there is no variation in the ‘treatment variable’ (Meyer and von Cramon-Taubadel, 2004, p. 588).

The present analysis goes beyond this research by comparing the cost pass-through rates for a large number of different brands and retailers. The empirical model is based on a comprehensive data set comprising a long panel of store-level retail prices and wholesale costs for up to 90 different brands of milk sold in 327 retail stores across Germany. The abundance of the data allows us to identify heterogeneity in pass-through across retailers and measure conditional correlations between pass-through rates and factors such as retailer market power, measures of consumer search and menu costs, and product characteristics.² In a two-stage procedure, we first estimate coefficients of the threshold-error-correction-process, which are then used as a dependent variable in a regression to measure conditional correlations with our measures of market power, menu- and search costs.

2. Literature

In terms of the empirical approach applied, the present analysis is closely related to studies in the retail gasoline market. Deltas (2008), Verlinda (2008) and Lewis (2011) investigate pass-through rates in multiple geographic markets and across many sellers and aim at examining observed differences between markets and regions with station and local market level characteristics. According to Meyer and von Cramon-Taubadel (2004) two explanations for an *aCPT* are predominantly used in the existing literature: non-competitive markets (market power) and menu costs. The first hypothesis states that (oligopoly) retailers may exert market power by raising prices more rapidly in response to a cost increase than by cutting prices in response to a cost decrease. Borenstein et al. (1997), for example, argue that ‘prices are sticky downward because when input prices fall the old output price offers a natural focal point for oligopolistic sellers’ (p. 324).³

The second hypothesis relies on costs related to adjusting quantities and prices. Menu costs imply a lagged response to cost shocks (Ball and Mankiw, 1994). Supply adjustment costs (including menu costs) can also explain asymmetries in price adjustment if these costs are asymmetric with respect to an increase or a decrease in output quantities and/or prices.⁴ Levy et al. (1998), however, cast some doubt on this asymmetry in menu costs: ‘given the structure of the price change process, the cost of labor used, the cost of preparing and delivering price tags, and the cost of in-store managerial time are not likely to be higher for price increases than price decreases’ (p. 113).⁵ The authors also explore one specific component of the costs of price adjustment – the so-called ‘mistake costs’ (i.e. costs associated with mistakes that occur in the price change process) – that may have an asymmetric effect on price adjustment. According to Levy et al. (1998), costs associated with these mistakes will be low if they favor the customers, but they can be very high if the mistakes favor the store, since the store may lose reputation and customers’ goodwill and may even face legal problems. The authors argue that ‘the asymmetric effects of the mistake cost component of the price adjustment costs may indeed deter price increases more often than price decreases’ (p. 113).

Retailers’ incentives to adjust prices asymmetrically to cost increases and decreases can also differ between products. Ward (1982) suggests that retailers of perishable products might hesitate to raise prices for fear of reduced sales leading to spoilage. This would lead to negative *aCPT*. Ward’s explanation is challenged by Heien (1980), who argues that ‘changing price is not a problem for perishables, but for items with a long shelf life, price changing is costly both in terms of time to put on new labels and in goodwill lost’ (p. 15).⁶ More specifically, *aCPT* can be the result of households’ unequal costs of maintaining relatively high or low inventories. For products that are more expensive to store in the household (perishable products), the costs of experiencing a stock-out at the retailer will be particularly high. To ward off a stock-out of these

² Previous investigations of the milk market have found evidence of asymmetric pass-through. Kinnucan and Forker (1987) observe pricing asymmetries in retail prices of dairy products (milk, cheese, butter, and ice cream) in the US, with larger and speedier reactions when farm prices are increasing. Serra and Goodwin (2003) find evidence for *aCPT* in dairy products in Spain. Chavas and Mehta (2004) analyze the butter market in the US for the period from 1980 to 2001. They find strong support for asymmetry in the adjustment of retail prices, with a stronger reaction when confronting wholesale price increases compared to wholesale price decreases. Fernandez-Amador et al. (2010) observe significant asymmetries in the vertical price transmission mechanism between producer and consumer prices of milk products in Austria using monthly data for the period from January 1996 to February 2010.

³ A reviewer correctly pointed out that high markups are not necessarily associated with collusion. Monopoly and nearly monopoly markets also have high markups, but may not indicate asymmetric cost pass-through.

⁴ In the Ball and Mankiw (1994) model, positive inflation rates can explain an asymmetric price transmission. Firms increase prices to correct for accumulated and anticipated inflation. Transmission of negative shocks would be less necessary as inflation would already have adjusted the prices. This model, however, does not explain any differences in *aCPT* between products within an economy.

⁵ He et al. (2013) argue that the form of the pricing asymmetry in a model with menu costs critically depends on the shape of the consumer demand curve. If the demand function is convex, a firm is more likely to change its price in response to a negative cost shock and less likely to change its price responding to a positive cost shock. A concave demand function has the opposite effect. Similarly, Meyer and von Cramon-Taubadel (2004) conclude: ‘In summary, . . . attempts to explain APT [asymmetric price transmission] based on adjustment costs lead to ambiguous and sometimes contradictory results, with some authors providing arguments for positive APT, and others for negative’ (p. 590).

⁶ Similarly, Rotemberg (2009, 2010) argues in a formal model, in which firms internalize the cost-related regret that consumers experience when prices change unexpectedly: ‘Price changes can trigger consumer regret. If a good is storable and people notice an increase in its price, they are likely to regret not having purchased earlier, while they regret not having waited if they see a price decline (Rotemberg, 2009, p. 1)’.

products, firms may lower their own prices at a slower rate than the decline in upstream prices. Thus, a cost reduction will be passed on to consumers more slowly than cost increases, which implies a positive asymmetry in cost pass-through rates for perishable products. For products that can be stored more easily in the household, the degree of asymmetry in cost pass-through rates should be smaller.⁷ Costs associated with product stock-outs can be substantial; Matsa (2011) argues that product availability is an important dimension of quality in the supermarket industry. According to the author, stock-outs are estimated to cost U.S. supermarket retailers \$6 billion to \$12 billion in immediate lost sales each year in addition to the potentially even greater risk that some consumers will also shift their future shopping to competing stores. To avoid these costs, supermarkets typically spend twice as much on inventory as on advertising.⁸

The importance of stock-out costs as an explanation for *aCPT*-rates has been challenged by the observation of rather slow price transmission rates for some products (in some cases, cost shocks take many weeks to pass through to retail prices). Given this slow speed of adjustment, one would expect re-stocking of these products to be rather easy during such a long time period. The present paper, however, does not suggest that stock-out costs fully explain slow price transmission rates. Our analysis rather aims at providing empirical evidence for differences in the speed and asymmetry of price transmission between products, which might be related to product-specific differences in stock-out costs: *aCPT*-rates should be larger for perishable products than for products that can be stored more easily.

More recent studies provide additional arguments for *aCPT* based on consumer search costs. Tappata (2009) argues that rational consumers will search less when prices are falling compared to when they are rising. Therefore, prices are slower to adjust in a downward direction. While Tappata assumes symmetric learning by consumers, Yang and Ye (2008) focus on asymmetric learning: 'When a positive cost shock occurs, all the searchers immediately learn the true state; the search intensity, and hence the prices, fully adjust in the next period. When a negative cost shock occurs, it takes longer for non-searchers to learn the true state, and the search intensity increases gradually, leading to slow falling of prices' (Yang and Ye, 2008, p. 547). In Cabral and Fishman (2012), firms experience heterogeneous cost shocks and consumers search sequentially. Consumers' willingness to search is low (high) when they observe small (large) price variations. Thus, firms' incentives to pass cost shocks to final prices differ depending on the size of these shocks. When cost changes are small, sellers increase prices moderately without losing customers but do not reduce prices. The opposite asymmetry occurs when the cost change is large. The following sections aim at examining cost pass-through rates empirically.

3. Market description, data and model specification

German dairies process about 20 per cent of the milk produced in the EU-27 (AMI, 2010). About one fourth of the German dairy companies are cooperatives, which are generally larger than non-cooperatives; their total share of processed milk is about 50 per cent. The market share of the six largest dairy companies is almost 50 per cent, and four of them are cooperatives. Dairy companies deal directly with food retailers. The food retailing sector is even more concentrated than the processing sector. In 2010, the top six grocery retailers provided more than 75 per cent of the sector's turnover. Among the different retail business formats (with respect to store size and store product assortment) discounters are particularly important; in 2009 discounters accounted for 56 per cent of milk retail sales in Germany. Consumers in Germany are highly price sensitive and dairy products are key items that influence consumers' store choices. Almost 75 per cent of the milk consumed in Germany are private labels. A more detailed description of the German milk processing and retailing market is available in Herrmann et al. (2009) and Loy et al. (2015).

The empirical analysis uses weekly retail prices of 90 different brands of milk (one-liter packs, with 3.5 percent fat) from 327 grocery stores for the period from 2005 to 2008. The data are obtained from Symphony IRI Group (SIG, 2011). Not all 90 brands are sold in all stores; in total 917 individual price series for 90 different brands and 327 grocery stores are available. Note that the price series include price promotions ('sales'), i.e. significant temporary price reductions that are unrelated to cost changes (Hosken and Reiffen, 2001). To prevent our price transmission estimates from being biased by the incidence of sales, we eliminate all price promotions from the retail price series and replace sales prices with the regular prices prior to the sale.⁹

The retail milk market is particularly suited for studying cost pass-through rates for several reasons. Milk is a frequently purchased product and an important category for both consumers and retailers. In periods of high retail prices as well as during sporadic outbreaks of price warfare, milk prices have been the subject of continuing interest to retailers and dairy farmers, as well as to consumers and government agencies. Secondly, due to the short shelf life and high perishability of milk, it is difficult for food processors, retailers and consumers to stockpile this product and hold large inventories in periods

⁷ Slonim and Garbarino (2009) show that the direction and strength of the asymmetry depend on the product's stockability (i.e., the number of units of the product a consumer will carry) and the strength of the reference price (loss aversion) effects.

⁸ The issue of stock-outs has begun to receive increased interest in the business literature as researchers have started to examine the short- and long-run consequences of stock-outs on consumers (see Corsten and Gruen (2004) and Musalem et al. (2010) among others). In a comprehensive study on the incidence and consequences of stock-outs across different product categories and geographies, Gruen et al. (2002) observe that the average out-of-stock rate in the United States and in Europe is about 8% at any point in time and that the costs associated with out-of-stocks can be substantial for some product categories.

⁹ We also checked the impact of excluding promotional prices by running models with the original series. The results show only small differences, the main conclusions remain unchanged. Results are available from the authors upon request.

of low prices. While the missing information on retailer inventories is a major concern for many empirical studies of cost pass-through rates, retailer inventories are less likely to be an important issue with milk. Finally, in contrast to most retail settings, where wholesale costs of retailers are unobservable to researchers, wholesale prices for milk can be measured very accurately. In the present case, we obtain wholesale prices for milk from an extensive survey of the German dairy industry conducted by the BLE ('Bundesanstalt für Landwirtschaft und Ernährung'). The BLE collects the data for each dairy (including the costs of transportation to the retailer). Due to confidentiality reasons, BLE only releases the average wholesale prices over all dairies. However, we interviewed dairy managers who confirmed that differences between retailers (due to different transportation costs, for example) are negligible. The pasteurization and homogenization involved in processing raw milk is highly standardized, with negligible differences across firms and locations, which justifies using the same wholesale price for each retailer and each brand. The market experts interviewed (CEOs of major dairy companies in the German market) confirmed that our cost measure is representative (with only minor differences of a few Eurocent per liter) for all dairies.

For each price series (brand of milk sold at one retailer), we estimate a restricted bivariate three-regime-threshold-error-correction-model, which allows asymmetric price responses. The estimation procedure considers the proper lag length, tests for threshold co-integration and the number of thresholds (for details see Loy et al., 2015). Eq. (1) shows the cost pass-through model.

$$\Delta p_t^R = \alpha_0 + \delta^- I_t^1 \mu_{t-1} + \delta^0 I_t^2 \mu_{t-1} + \delta^+ I_t^3 \mu_{t-1} + \sum_{j=1}^k \alpha_j \Delta p_{t-j}^R + \sum_{j=1}^k \beta_j \Delta p_{t-j}^W + \nu_t \quad (1)$$

Changes of the retail price (Δp_t^R) depend on changes of the wholesale price (Δp_t^W) and deviations from the long-run price-cost-equilibrium (μ_{t-1}), which are separated by indicator variables (I_t^1, I_t^2, I_t^3) into three regimes.¹⁰ By estimating Eq. (1) for each individual product (brand of milk sold at a specific store), we obtain estimates for lower and upper regime error-corrections δ^- ; δ^+ . From the parameter estimates δ^- and δ^+ , we can derive a natural measure for the speed of the cost pass-through ($sCPT$) as well as for the asymmetry of the cost pass-through ($aCPT$), which are calculated as follows: $CPT = |\delta^+ + \delta^-|/2$ and $CPT = \delta^+ - \delta^-$.

In a second step, we investigate the relationship between differences in $sCPT$ and $aCPT$ between the 917 products (brands sold in different stores) and our measures of market power, menu- and search costs. In this two-stage procedure, we estimate all coefficients of the CPT model for each individual retail price series separately. Note that this procedure does not take into account any form of interdependence that might exist between individual retail price series. As an alternative, one could also estimate the model in a one-step panel model. However, a panel model places strong restrictions on individual CPT processes: all CPT model parameters are assumed identical for all individual series. The efficiency gain of the panel model comes at the cost of placing model parameter restrictions. We decided on the most flexible form, in particular because the number of time-related observations of our sample is quite large ($t = 208$).

We follow previous studies (Connor and Peterson, 1992; Barsky et al., 2003) by using the relative markup between average individual retail and wholesale prices (i.e. the Lerner Index LI) as our measure of market power. Note that this margin does not cover marginal costs of retailing; our measure thus represents an upper bound for retailers' markup on marginal costs.¹¹ Our measure of menu costs (MC) is obtained from estimation results of the threshold-error-correction model. The inner regime of the threshold-error-correction-mechanism captures smaller deviations from the long-run cost-price-equilibrium. Due to adjustment costs these small deviations need to cumulate over time before they are passed on to retail prices. Over the range of the inner regime, retail prices may not change at all. A greater range, thus, indicates higher menu costs. We thus define $MC = \theta^+ - \theta^-$.

While search models have played an important role in economics ever since the seminal article by Stigler (1961), empirical evidence on the importance of search costs is scarce due to the obvious difficulties in measuring these costs.¹² In the context of multi-product retailers, a distinction between cross-store and cross-product search costs is important. Richards et al. (2014), one of the very few studies investigating the implications of search costs for price adjustment empirically, focus on cross-product (within store) search costs. The authors assume that consumer search costs vary directly with the number of products in a retail store. Greater variety of products implies higher search costs because consumers must evaluate a greater number of alternatives before finding the one they prefer. However, as pointed out by one of our referees, the comparison of different product prices within a given store will typically be easier than comparing prices across stores. If consumers are willing to switch between stores to purchase the cheapest product, it might be more important (but also more costly) to obtain information about what other stores are charging (cross-store search costs). Unfortunately,

¹⁰ Two thresholds (θ^- , θ^+) are estimated following the procedure by Chan (1993). The indicator variables are defined according to the regimes separated by the thresholds: if $\mu_{t-1} < \theta^-$, then $I_t^1 = 1$, else $I_t^1 = 0$. If $\mu_{t-1} > \theta^+$, then $I_t^3 = 1$ else $I_t^3 = 0$ and $I_t^2 = 1 - I_t^1 - I_t^3$.

¹¹ Connor and Peterson (1992) and Barsky et al. (2003) measure market power in a similar way by calculating the difference between national brand and private label prices.

¹² Some authors (Brown and Goolsbee (2002) and Tang et al. (2010), for example) have attempted to understand the nature of search costs by comparing the degree of price dispersion between online and traditional retail markets, for example. Since milk is not sold online, this measure cannot be used in the present context.

Table 1

Summary statistics of estimation results from first stage regression.

Parameters and tests	Mean	Std. dev.	Share of estimates significantly different from zero at 5% level
	(1)	(2)	(3)
δ^-	−0.316	0.213	93%
δ^0	−0.045	0.270	8%
δ^+	−0.118	0.103	62%
$\delta^+ - \delta^-$	0.198	0.161	76%
θ^-	−0.035	0.023	n.a.
θ^+	0.058	0.034	n.a.

Own calculations based on SIG (2011) using Stata (2011).

The thresholds θ^- and θ^+ are estimated following Chan's (1993) procedure. To our knowledge, no formal test on the size of thresholds is available so far.

a good proxy for cross-store search costs is not available. Thus, in the present analysis, we follow Richards et al. (2014) and use the number of different milk brands in a store as our only measure of consumer search costs (SC).

We also control for the type of milk (fresh versus processed at ‘ultra-high temperature’), the organizational structure of the food processor (cooperative versus non-cooperative), the type of the brand (private label versus national brand), and the type of the retail channel (discounter versus non-discounter). These variables are included in the matrix $\mathbf{X}$ in Eq. (2).

$$\begin{aligned} sCPT_i &= \alpha_0 + \alpha_1 LI_i + \alpha_2 MC_i + \alpha_3 SC_i + \mathbf{X}_i \boldsymbol{\gamma} + \varepsilon_i \\ sCPT_i &= \beta_0 + \beta_1 LI_i + \beta_2 MC_i + \beta_3 SC_i + \mathbf{X}_i \boldsymbol{\delta} + \varepsilon_i \end{aligned} \quad (2)$$

In Eq. (2), subscript i refers to the different products (brands in different stores), $\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1, \beta_2, \boldsymbol{\gamma}$ and $\boldsymbol{\delta}$ and are parameters to be estimated and ε as well as ε are IID error terms.

4. Results and conclusions

Time series unit root, co-integration, and threshold co-integration tests are applied to each of the 917 price series. Furthermore, we use panel tests for unit roots and co-integration. Granger-causality tests suggest that wholesale prices are exogenous. All procedures indicate non-stationary behavior of price series and co-integration between wholesale and retail prices. The cost pass-through processes indicate non-linear behavior which is modeled by a threshold-error-correction-mechanism. The number of thresholds is determined using the procedure developed by Strikholm and Teräsvirta (2006). For 90 per cent of all retail price series for milk, the procedure suggests at least one threshold. In 49 per cent of all cases, the assumption of one threshold is rejected against the two-threshold model (i.e. three regimes). We thus estimate the two-threshold model (with three regimes) for all price series since this is the more flexible specification.

Nearly all (99 percent) of the steady state price-cost-equilibria lie within the range of thresholds $[\theta^+, \theta^-]$. On average, 150 out of 208 observations (75 percent) refer to the inner regime. The lower threshold is on average −3.5 Eurocent per liter; the upper one is 5.8 Eurocent per liter. The adjustment coefficient estimates (δ^0) for the inner regime ($\theta^- \leq \mu_{t-1} \leq \theta^+$) are small, the average is −0.04. The parameter estimates for δ^0 are not significantly different from zero in 92 percent of cases. In the vast majority of cases, adjustment coefficients for the outer regimes (δ^- ; δ^+) are absolutely larger than the coefficients for the respective inner regime. The estimates for the lower regimes (δ^- for $\mu_{t-1} \leq \theta^-$) are larger in absolute terms than the ones for the upper regime (δ^+ for $\mu_{t-1} \geq \theta^+$). The average rate of decay for a margin that is smaller than the lower threshold (δ^-) is −0.32 compared to −0.12 for δ^+ . Furthermore, the parameter estimates for δ^- are significantly different from zero in nine out of ten cases, while the estimates for δ^+ are significantly different from zero in 62 percent of all cases. Table 1 reports descriptive statistics of the first-stage estimation results.

In a second step, we investigate the conditional correlation between CPT-rates (the speed $sCPT = |\delta^+ + \delta^-|/2$ as well as the asymmetry $aCPT = \delta^+ - \delta^-$ of cost pass-through rates) and measures of market power, menu- and search costs.¹³ Table 2 reports descriptive statistics of all variables used in Eq. (2), conditional correlations of the speed and asymmetry of price transmission with market, store, and product characteristics are shown in Table 3.

Following Lewis and Linzer (2005) we apply OLS with robust standard errors (in columns (1) and (3) of Table 3) and WLS. Note that the empirical analysis might be affected by problems associated with the endogeneity of explanatory variables. Firstly, an asymmetric CPT might mechanically generate a higher Lerner Index: if the prices of some brands adjust more quickly to cost increases than to cost decreases, the average mark-up for these brands will be larger. Secondly, the same price data is used in both, our definition of the dependent variables ($sCPT$ and $aCPT$) and in our measure of menu costs (MC), which might cause difficulties with regard to identifying a causal link. To consider the potential problem of endogeneity of the Lerner Index (LI) and our measure of menu costs (MC), we employ the instrumental variable approach suggested by Lewbel

¹³ The parameter estimates suggest positive asymmetries for most products (brands in different stores). We exclude estimates that indicate no asymmetries or negative ones. This reduces the sample by 7 percent; the number of observations for the second step is reduced to 857. The exclusion of these estimates does not change the main results of our empirical analysis.

Table 2
Descriptive statistics.

	Obs.	Mean	Std. dev.	Min.	Max.
Asymmetry (<i>aCPT</i>)	857	0.219	0.145	0.003	0.828
Speed (<i>sCPT</i>)	857	0.223	0.147	−0.000	0.607
Lerner Index (<i>LI</i>)	857	0.340	0.124	0.086	0.538
Menu costs (<i>MC</i>)	857	0.094	0.045	0.001	0.288
Search costs (<i>SC</i>)	857	7.540	2.863	2	14
Non-cooperative dairy (<i>P</i>)	857	0.360	0.480	0	1
Private label (<i>PL</i>)	857	0.331	0.470	0	1
UHT-milk (<i>UHT</i>)	857	0.669	0.470	0	1
Discounter (<i>DC</i>)	857	0.076	0.265	0	1

Source: own calculations based on [SIG \(2011\)](#) using Stata (2011).

Asymmetry = Difference of error correction response ($\delta^+ - \delta^-$); Speed = Average speed error correction response ($(|\delta^+ + \delta^-|)/2$); Lerner Index = ($\emptyset$ retail price – $\emptyset$ wholesale price)/ $\emptyset$ retail price; Menu costs = Range of thresholds (range of inner regime); Search costs = # stock-keeping units. The number of observations is 857.

Table 3
Estimation results of second-stage estimation model (dependent variables are the speed and the asymmetry of cost pass-through rate).

Estimation method	Dependent variable is $CPT \equiv \frac{ \delta^+ + \delta^- }{2}$				Dependent variable is $CPT \equiv \delta^+ - \delta^-$			
	Parameter value	(t-ratio)	Parameter value	(t-ratio)	Parameter value	(t-ratio)	Parameter value	(t-ratio)
	OLS (1)		IVreg (Lewbel, 2012) (2)		OLS (3)		IVreg (Lewbel, 2012) (4)	
Explanatory variables								
Constant	0.491	(16.31)***	0.503	(13.95)***	0.507	(15.09)***	0.513	(12.15)***
Lerner Index (<i>LI</i>)	−0.803	(−11.54)***	−0.756	(−8.56)***	−0.638	(−7.77)***	−0.619	(−5.52)***
Menu costs (<i>MC</i>)	0.008	(0.13)	−0.274	(−2.25)**	−0.317	(−3.33)***	−0.437	(−2.86)**
Search costs (<i>SC</i>)	−0.004	(−3.97)***	−0.004	(−3.92)***	−0.003	(−1.84)*	−0.003	(−1.83)*
Non-cooperative dairy (<i>P</i>)	0.050	(6.49)***	0.046	(5.84)***	−0.006	(−0.49)	−0.008	(−0.58)
Private label (<i>PL</i>)	0.082	(4.75)***	0.080	(3.95)***	−0.003	(−0.17)	−0.005	(−0.22)
UHT-milk (<i>UHT</i>)	−0.015	(−2.31)**	−0.014	(−2.21)**	−0.031	(−3.56)***	−0.031	(3.58)***
Discounter (<i>DC</i>)	0.012	(0.97)	0.016	(1.09)	0.035	(2.15)**	0.039	(2.37)**
R-squared	0.705		0.699		0.396		0.396	
Adj. R-squared	0.703		0.697		0.391		0.390	

Source: own calculations based on [SIG \(2011\)](#) using Stata (2011).

We report [White's \(1980\)](#) heteroscedasticity-consistent standard errors. Asterisks denote statistical significance in a t-test at 1 per cent (***), 5 per cent (**) or 10 per cent (*) levels. The number of observations is 857. The reference category is a national brand of fresh milk, produced at a cooperative dairy, sold at non-discounters in Central Germany. Estimates for the WLS procedure are available from the authors upon request. The Hausman-test shows that both estimation procedures result in the same set of estimators. For *sCPT* the χ^2 (7) = 11.55 (*p*-value = 0.12) and for *aCPT* the χ^2 (7) = 1.22 (*p*-value = 0.99).

(2012) in columns (2) and (4).¹⁴ [Table 3](#) suggests that the parameter estimates between the OLS and the IV approach are very similar. In fact, employing a Hausman test does not reject the same set of estimates for OLS and Lewbel's instrumental variable approach. Thus, endogeneity appears not to be an issue here.

As an additional robustness test, we re-estimate the model with store-fixed effects. Two explanatory variables that are fixed for each retailer (our measure of search costs, i.e. the number of products in each store, as well as the type of the retail channel, i.e. discounter versus non-discounter), had to be removed in the fixed-effects model. Since the parameter estimates for all other variables are hardly affected, they are not shown in [Table 3](#) but are reported in [Table A1](#) in [Appendix A](#).

[Table 3](#) reports a negative parameter estimate for the Lerner Index (*LI*) which is significantly different from zero at the 99%-level in all four estimation models. Columns 1 and 2 suggest a negative correlation between price-cost margins and our measures of the speed and asymmetry of price adjustment. This finding is in line with the theoretical arguments and the empirical evidence (for the gasoline market) reported in [Borenstein and Shepard \(2002\)](#), for example. The estimation results reported in columns (3) and (4) suggest that the (positive) asymmetry between the adjustment coefficients of the outer regimes δ^- , δ^+ is associated with lower values of the Lerner Index (*LI*). A causal interpretation of these estimates is that retailers adjust prices of products (brands in individual stores) more quickly to cost increases than to cost decreases where the markup on wholesale prices is small. For products with a larger markup on wholesale prices, the asymmetry of price adjustments with respect to cost shocks is significantly smaller. Note, however, that we cannot fully rule out the existence of an omitted variable that leads to both, high margins and low values for the asymmetry of price transmission.

The parameter estimate for our measure of the magnitude of menu costs (*MC*) is negative and significantly different from zero in columns (2), (3) and (4). Larger menu costs are associated with a slower speed of price adjustment (column (2)). Column (3) and (4) suggest that the positive asymmetry in cost pass-through rates tends to be smaller in markets where

¹⁴ More details are provided in [Appendix A](#).

menu costs are important. This finding is consistent with the existence of ‘mistake costs’ – although not a formal test of it – since this type of price adjustment costs would indeed deter price increases more often than price decreases (Levy et al., 1998).

Some of the effects of menu-costs might also be picked up by product-, store- and brand dummies included in our empirical model (see below). Levy et al. (1997) use store-level data to directly measure menu costs at five multistore supermarket chains and observe substantial differences in menu costs between chains. In an additional estimation experiment (available from the authors upon request), we estimate the relationship between the range of the thresholds (our measure of menu costs) and dummies for non-cooperative dairies, private labels, and the retail channel (discounter). For the private label dummy, we find a significant negative parameter estimate indicating that the inner regime (the regime with almost no cost-price-adjustment) is smaller for private labels than for national brands. This could be interpreted as an indication for lower menu costs for private labels (we are grateful to one of the referees for bringing up this issue).

Our measure of consumer search costs (SC), i.e. the number of different milk brands offered in the retail store, is negatively correlated with the speed of price adjustment to cost changes: prices adjust more sluggishly to cost changes in stores with many different brands (high consumer search costs). The impact of SC is significantly different from zero at the 10% level in columns (3) and (4). Our results thus do not provide empirical support to the recent theoretical literature on asymmetric cost pass-through, which suggests a positive relationship between search costs and pricing asymmetries. However, the present analysis is based on measuring cross-product (within store) search costs only. The question whether (potentially more important) cross-store search costs would exert the expected positive impact on pricing asymmetries cannot be answered based on the available data but is deferred to future research.

We further find that non-cooperative brands and private labels adjust prices more quickly to cost changes than cooperative and/or national brands (columns 1 and 2 in Table 3). No significant differences in the asymmetry of cost pass-through rates for these variables are reported in Table 3. We find a higher speed (columns 1 and 2) but less asymmetry (columns 3 and 4) in the price adjustment to cost changes for ‘ultra-high temperature’ (UHT) milk. Discounters do not pass on costs more quickly than other retailers, however, they tend to adjust more asymmetrically: the value of *aCPT* is significantly higher for discounters than for all other retailers in columns (3) and (4).

In summary, our results confirm Peltzman’s (2000) argument which suggests that ‘... attributing asymmetries to imperfect competition is unlikely to be rewarding’ (p. 468). In contrast to much of the existing literature, which explains the ‘rockets and feathers’ phenomenon as an apparent result of retailers exercising market power, we find the opposite result: the degree of asymmetry in prices is negatively related to market power (decreases with the Lerner-Index). If we assume an oligopoly for the market under study, this result implies a rejection of the theory of Borenstein et al. (1997); or the market under study is not an oligopoly but e.g. a monopoly or nearly monopoly market, which also have high markups without perhaps an asymmetric cost pass-through.¹⁵

Furthermore, we find that our measure of (cross-product) search costs does not account for observable differences in *aCPT*-rates between products. Whether asymmetries in price transmission are due to consumers’ cross-store search costs should be analyzed in future research.

The ‘rockets and feathers’ observation is better explained by differences in menu costs (‘mistake costs’) as well as costs associated with stock-outs. Note that in both cases, an asymmetry in consumer learning is involved where consumers react asymmetrically to satisfying and unsatisfying experiences. In the case of stock-outs, for example, Gaur and Park (2007) argue that “in practice, customers do react substantially and negative to poor service (e.g., stock-outs), which may lead them to switch retailers on subsequent trips” (p. 227). The authors further suggest that stock-out costs will differ between product categories: “For a necessity item such as milk, for instance, consumers may react more strongly to a stock-out than to a satisfying visit because they might take it for granted that such items should always be in stock. For a search item such as fashion clothing, on the other hand, consumers may weigh a satisfying visit much more than a stock-out of one of the variants because they expect to search for these items” (p. 231–2). Similarly, if costs of experiencing a stock-out for consumers are larger for perishable products (i.e. fresh milk), retailers will adjust prices for these products more quickly upwards than downwards. Our finding of a significantly smaller degree of *aCPT* for UHT milk as opposed to fresh milk is consistent with—although not a formal econometric test of—this argument.¹⁶ Empirical work by Corsten and Gruen (2004), Musalem et al. (2010), and Matsa (2011) suggests that costs associated with stock-outs indeed are substantial.

Data permitting, it would be interesting to measure and control for the impact of perishability, the stock of inventories and the importance of cross-store search costs directly. Explicitly analyzing inventory policies of retailers for different products as well as their implications for asymmetries in cost pass-through rates should prove conducive to enhancing our understanding of retailer strategies and providing implications for managerial decision-making at the retailer level.

¹⁵ Either way, the clear negative association between the LI and the *aCPT* shown in the data calls for an extension of the theory. However, such an extension goes beyond the scope of this paper.

¹⁶ Note that this explanation for *aCPT*-rates will be particularly relevant for retailers who sell a larger share of their milk products to smaller businesses (restaurants, cafes or other businesses not large enough to buy at wholesale). These customers will have the largest incentive to carry sufficient inventory of storable products. While some of the existing empirical studies on the incidence and consequences of stock-outs found significant differences between product groups (Gruen et al. (2002) and Corsten and Gruen (2004)), systematic empirical evidence for (different types of) milk products is not available so far.

Acknowledgments

Jens-Peter Loy gratefully acknowledges the financial and other support by the Vienna University of Economics and Business during a research visit in March 2014.

Appendix A.

(a) Results of fixed-effects estimation

Table A1

Fixed-effects estimation results of second-stage estimation model (dependent variables are the speed and the asymmetry of cost pass-through rate).

	Dependent variable is $CPT \equiv \frac{ \delta^+ + \delta^- }{2}$		Dependent variable is $CPT \equiv \delta^+ - \delta^-$	
	Parameter value	(t-ratio)	Parameter value	(t-ratio)
	(1)		(2)	
Explanatory variables				
Constant	0.439	(16.65)***	0.473	(12.73)***
Lerner Index (<i>LI</i>)	−0.768	(−11.14)***	−0.634	(−6.52)***
Menu costs (<i>MC</i>)	0.069	(0.85)	−0.194	(−1.69)*
Non-cooperative dairy (<i>P</i>)	0.050	(5.25)***	−0.005	(−0.39)
Private label (<i>PL</i>)	0.096	(6.29)***	0.008	(0.40)
UHT-milk (<i>UHT</i>)	−0.017	(−2.64)***	−0.033	(−3.69)***
R-squared (within)	0.735		0.432	
R-squared (between)	0.664		0.364	
R-squared (overall)	0.696		0.387	

Source: own calculations based on SIG (2011) using Stata (2011).

We report White's (1980) heteroscedasticity-consistent standard errors. Asterisks denote statistical significance in a *t*-test at 1 per cent (***), 5 per cent (**) or 10 per cent (*) levels. The number of observations is 857. The reference category is a national brand of fresh milk produced at a cooperative dairy.

(b) Instrumental variable approach

Conventionally, identification of the structural parameters is obtained by assuming exclusion restrictions, or equivalently by postulating the availability of instrumental variables. Since the exclusion restrictions may not always be justified, the present analysis follows Lewbel (2012), who relies on heteroscedasticity in the errors to achieve identification. The main advantages of this method are that no external instrumental variables are necessary to identify the structural parameters and that the associated estimator usually takes the standard form of the generalized method of moments (GMM). "For each exogenous variable an instrument is constructed from $Z_j = (X_j - \bar{X}) \cdot \varepsilon_j$, where ε is a vector of residuals from the first stage regression of each endogenous regressor on all exogenous regressors, including a constant vector. These auxiliary regression residuals have zero covariance with each of the regressors used to construct them, implying that the means of the generated instruments will be zero by construction. However, their element-wise products with the centered regressors will not be zero, and will contain sizable elements if there is clear evidence of 'scale heteroskedasticity' with respect to the regressors" (Baum et al., 2013: 11–12).

Lewbel (2012) shows that identification can be achieved without imposing the standard exclusion restrictions when there are some exogenous variables *Z* in the structural equation and errors are heteroscedastic. In the present context, the *Z* vector comprises the following variables: *SC*, *PL*, *P*, *UHT*, and *DC* (see Table 2 or Table 3). Identification requires that the error terms in the first-stage regressions (for *LI* and *MC*) are heteroscedastic. Following Lewbel (2012), we use the Breusch–Pagan test of heteroscedasticity. The results show that the null of homoscedastic errors is clearly rejected in each case with a *p*-value equal to 0.01 or less. Sargan's test of overidentification is not rejected ($\alpha = 0.202$); thus, we employed proper instruments.

Table A2 shows results from the first-stage regression for the Lerner-Index (*LI*) for key variables in the *Z* vector. Non-cooperative dairies (*P*), which are more flexible in making strategic decisions and produce more and higher value national

Table A2

First-stage regression for the Lerner-Index (*LI*).

Explanatory variables	Parameter estimates	(t-ratio)
Non-cooperative dairy (<i>P</i>)	0.084	(20.02)***
Private label (<i>PL</i>)	−0.182	(−40.65)***
Ultra high heated milk (<i>UHT</i>)	0.001	(0.40)
Discounters (<i>DC</i>)	−0.023	(−3.53)***
Constant	0.370	(110.19)***
	<i>N</i> = 857	<i>R</i> ² = 0.85

Source: own calculations based on SIG (2011) using Stata (2011).

brands, are found to have higher markups. Private labels (PL) are store brands that are used to promote the store and to present a low price alternative to national brands; thus PL are cheaper (lower markups) than national brands. Ultra high heated milk (UHT) is more expensive to produce but at the same time the handling and storage are cheaper; we do not observe a significant relationship with the Lerner-Index. Discounters (DC) tend to offer a small product assortment and follow an EDLP (everyday low price) strategy with lower markups.

References

- AMI (Agrarmarkt Informations-Gesellschaft mbH), 2010. *AMI-Marktbilanz Milch 2010*. Bonn.
- Ball, L., Mankiw, N.G., 1994. Asymmetric price adjustment and economic fluctuations. *Econ. J.* 104 (March), 247–261.
- Barsky, R.B., Bergen, M., Dutta, S., Levy, D., 2003. What can the price gap between branded and private-label products tell us about markups? In: Feenstra, R.C., Shapiro, M.D. (Eds.), *Scanner Data and Price Indexes*. University of Chicago Press, pp. 165–228.
- Baum, C.F., Lewbel, A., Schaffer, M.E., Talavera, O., 2013. Instrumental Variables Estimation Using Heteroskedasticity-Based Instruments. German Stata Users Group Meeting, Potsdam (<http://www.stata.com/meeting/germany13/abstracts/materials/de13.baum.pdf>).
- Borenstein, S., Shepard, A., 2002. Sticky prices, inventories, and market power in wholesale gasoline markets. *RAND J. Econ.* 33 (1), 116–139.
- Borenstein, S., Cameron, A.C., Gilbert, R., 1997. Do gasoline prices respond asymmetrically to crude oil price changes? *Q. J. Econ.* 112 (1), 305–339.
- Brown, J.R., Goolsbee, A., 2002. Does the internet make markets more competitive? Evidence from the life insurance industry. *J. Polit. Econ.* 110 (3), 481–507.
- Cabral, L., Fishman, A., 2012. Business as usual: a consumer search theory of sticky prices and asymmetric price adjustment. *Int. J. Ind. Organ.* 30, 371–376.
- Chan, K.S., 1993. Consistency and limiting distribution of the least squares estimator of a threshold autoregressive model. *Ann. Stat.* 21 (1), 520–533.
- Chavas, J.-P., Mehta, A., 2004. Price dynamics in a vertical sector: the case of butter. *Am. J. Agric. Econ.* 86, 1078–1093.
- Connor, J.M., Peterson, E.B., 1992. Market-structure determinants of national brand-private label price differences of manufactured food products. *J. Ind. Econ.* 40 (2), 157–171.
- Corsten, D.S., Gruen, T., 2004. Stock-outs cause walkouts. *Harv. Bus. Rev.* 82 (5), 26–28.
- Deltas, G., 2008. Retail gasoline price dynamics and local market power. *J. Ind. Econ.* 56 (3), 613–628.
- Eckert, A., 2013. Empirical studies of gasoline retailing: a guide to the literature. *J. Econ. Surv.* 27 (1), 140–166.
- Fernandez-Amador, O., Baumgartner, J., Crespo-Cuaresma, J., 2010. Milking the prices: the role of asymmetries in the price transmission mechanism for milk products in Austria. Working Paper in Economics and Statistics. University of Innsbruck, pp. 2010–2021.
- Frey, G., Manera, M., 2007. Models of asymmetric price transmission. *J. Econ. Surv.* 21 (2), 349–415.
- Gaur, V., Park, Y.-H., 2007. Asymmetric consumer learning and inventory competition. *Manag. Sci.* 53 (2), 227–240.
- Gruen, T.W., Corsten, D.S., Bharadwaj, S., 2002. Retail Out-of-Stocks: A Worldwide Examination of Extent, Causes and Consumer Responses. Grocery Manufacturers of America, Washington, DC.
- He, G., LaFrance, J.T., Perloff, J.M., 2013. Do supermarkets adjust prices asymmetrically? Working Paper. University of Berkeley, Available at: <http://are.berkeley.edu/candidate/sites/are.berkeley.edu/files/HeLaFrancePerloffPassThrough.pdf> (downloaded April 16, 2014).
- Heien, D.M., 1980. Markup pricing in a dynamic model of the food industry. *Am. J. Agric. Econ.* 62 (1), 10–18.
- Herrmann, R., Moeser, A., Weber, S., 2009. Grocery retailing in Germany: situation, development and pricing strategies. Working Paper No 41. Zentrum für internationale Entwicklungs- und Umweltforschung der Justus-Liebig-Universität Gießen, Gießen, Germany.
- Hosken, D., Reiffen, D., 2001. Multiproduct retailers and the sale phenomenon. *Agribusiness* 17 (1), 115–137.
- Kinnucan, H.W., Forker, O.D., 1987. Asymmetry in farm-retail price transmission for major dairy products. *Am. J. Agric. Econ.* 69, 285–292.
- Levy, D., Bergen, M., Dutta, S., Venable, R., 1997. The magnitude of menu costs: direct evidence from large U.S. supermarket chains. *Q. J. Econ.* 112 (3), 791–825.
- Levy, D., Dutta, S., Bergen, M., Venable, R., 1998. Price adjustment at multiproduct retailers. *Manag. Decis. Econ.* 19, 81–120.
- Lewbel, A., 2012. Using heteroscedasticity to identify and estimate mis-measured and endogenous regressor models. *J. Bus. Econ. Stat.* 30 (1), 67–80.
- Lewis, J.B., Linzer, D.A., 2005. Estimating regression models in which the dependent variable is based on estimates. *Polit. Anal.* 13 (4), 345–364.
- Lewis, M., 2011. Asymmetric price adjustment and consumer search: an examination of the retail gasoline market. *J. Econ. Manag. Strateg.* 20 (2), 409–449.
- Loy, J.-P., Holm, T., Steinhagen, C., Glauben, T., 2015. Cost pass-through in differentiated product markets: a disaggregated study for milk and butter. *Eur. Rev. Agric. Econ.* 42 (3), 441–471.
- Matsa, D.A., 2011. Competition and product quality in the supermarket industry. *Q. J. Econ.* 126, 1539–1591.
- Meyer, J., von Cramon-Taubadel, S., 2004. Asymmetric price transmission. *J. Agric. Econ.* 55 (3), 581–611.
- Misra, K.R., Khan, V., 2010. Asymmetries and dynamics of cost pass through in the US milk market. Working Paper. London Business School, Available at: <http://faculty.london.edu/kmisra/assets/documents/Cost.Passthrough.pdf>, downloaded 06.03.2014.
- Musale, A., Olivares, M., Bradlow, E.T., Terwiesch, C., Corsten, D., 2010. Structural estimation of the effect of out-of-stocks. *Manag. Sci.* 56 (7), 1180–1197.
- Perdiguer-Garcia, J., 2013. Symmetric or asymmetric oil prices? A meta-analysis approach. *Energy Policy* 57, 389–397.
- Peltzman, S., 2000. Prices rise faster than they fall. *J. Polit. Econ.* 108 (3), 466–502.
- Richards, T.J., Gómez, M.I., Lee, J., 2014. Pass-through and consumer search: an empirical analysis. *Am. J. Agric. Econ.* 96 (4), 1049–1069.
- Rotemberg, J., 2010. Altruistic dynamic pricing with customer regret. *Scand. J. Econ.* 112 (4), 646–672.
- Rotemberg, J., <http://www.nber.org/papers/w14933> 2009. Altruistic dynamic pricing with customer regret. Working Paper 14933. National Bureau of Economic Research, 1050 Massachusetts Avenue, Cambridge, MA 02138.
- Serra, T., Goodwin, B.K., 2003. Price transmission and asymmetric adjustment in the Spanish dairy sector. *Appl. Econ.* 35, 1889–1899.
- SIG: SymphonyIRI Group, 2011. Retail Scanner Data. SymphonyIRI Group GmbH, Düsseldorf.
- Slonim, R., Garbarino, E., 2009. Similarities and differences between stockpiling and reference effects. *Manag. Decis. Econ.* 30, 351–371.
- Stigler, G.J., 1961. The economics of information. *J. Polit. Econ.* 69 (3), 213–225.
- Strikholm, B., Teräsvirta, T., 2006. A sequential procedure for determining the number of regimes in a threshold autoregressive model. *Econom. J.* 9 (3), 472–491.
- Tang, Z., Smith, M.D., Montgomery, A., 2010. The impact of shopbot use on prices and price dispersion: Evidence from online book retailing. *Int. J. Ind. Organ.* 28, 579–590.
- Tappata, M., 2009. Rockets and feathers: understanding asymmetric pricing. *RAND J. Econ.* 40, 673–687.
- Verlinda, J.A., 2008. Do rockets rise faster and feathers fall slower in an atmosphere of local market power? Evidence from the retail gasoline market. *J. Ind. Econ.* 56 (3), 581–612.
- Ward, R.W., 1982. Asymmetry in retail, wholesale, and shipping point pricing for fresh vegetables. *Am. J. Agric. Econ.* 64 (2), 205–212.
- White, H., 1980. A heteroscedasticity consistent covariance matrix estimator and a direct test for heteroscedasticity. *Econometrica* 48, 817–838.
- Yang, H., Ye, L., 2008. Search with learning: understanding asymmetric price adjustments. *RAND J. Econ.* 39, 547–564.